Project Proposal

StudyBuddy – iOS Study Planner Application

Project Title: StudyBuddy – Smart Study Planner iOS Application
Submitted by: Isharat Jahan
Contact Email: isharat.isharatjahan@myokanagan.cuchd.in
Date Submitted:

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1. Introduction

This proposal presents the development of **StudyBuddy**, an iOS mobile application designed to help students efficiently manage their academic workload. The app integrates multiple study tools such as task management, timetable scheduling, focus tracking, and performance analytics into a single platform.

Students often struggle to organize their studies across different tools, leading to missed deadlines and inefficient study habits. StudyBuddy aims to solve this problem by providing a centralized, intuitive solution that enhances productivity and consistency.

The application will be developed using **Swift programming language** in **Xcode**, utilizing **SwiftUI** for interface design and **SwiftData** for local data storage.

This project is developed as part of an iOS Application Development course and demonstrates concepts such as modern UI design, data persistence, state management, and mobile app architecture.

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2. Understanding of the Project

The goal of this project is to design and implement a comprehensive **study management application** that enables students to organize and track their academic activities effectively.

The application will provide:

- A subject management system
- A customizable timetable with weekly and A/B rotation support
- Task tracking with deadlines and reminders

- A focus timer using the Pomodoro technique
- Study statistics and performance analytics
- Notes and grade tracking per subject

The system will store all data locally using SwiftData and present it through a responsive and user-friendly interface.

This project demonstrates the use of modern mobile development concepts including **MVVM architecture**, **data binding**, **local database management**, and **interactive UI design**.

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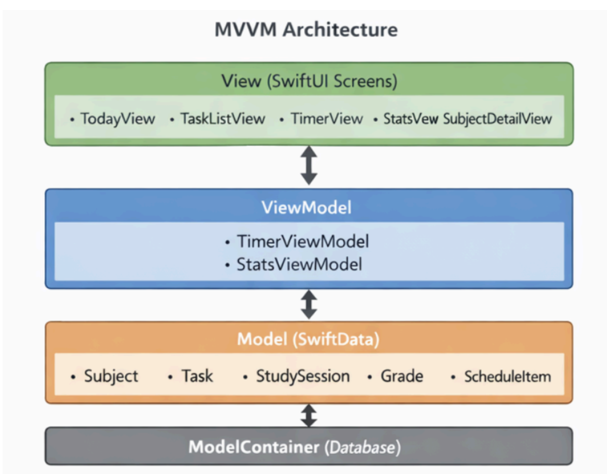
3. Proposed Scope of Work

a. Application Design and Setup

- Install and configure Xcode and Swift environment
- Design application using **MVVM architecture**
- Set up project structure and navigation flow
- Configure SwiftData model container



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b. User Interface Development

- Build UI using **SwiftUI**
- Implement screens including:
    - Today Dashboard
    - Task Manager
    - Timer Screen
    - Stats Dashboard

    ○ Subject Detail View
- Create a clean, modern, and responsive interface
- Add animations for better user experience

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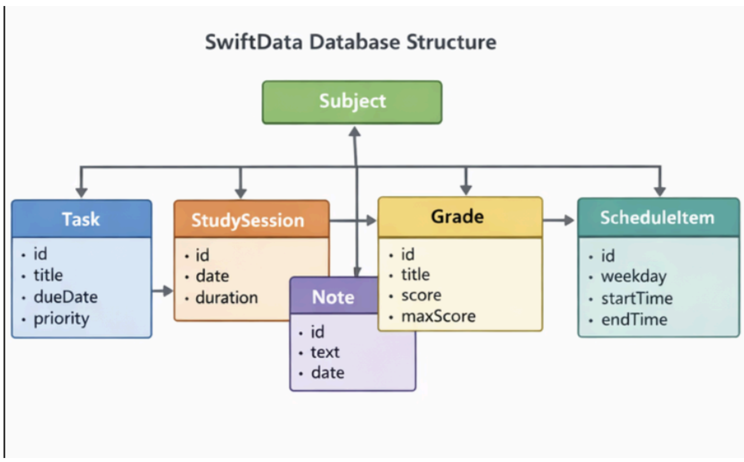
c. Data Management (SwiftData)

- Implement relationships with cascade delete
- Perform CRUD operations (add, delete, update)
- Ensure automatic UI updates using @Query



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d. Task and Schedule Management

- Allow users to create, edit, and delete tasks
- Add due dates, priorities, and reminders
- Implement timetable with weekday and A/B week rotation
- Display daily schedule dynamically

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e. Focus Timer System

- Implement Pomodoro timer:
    - Focus sessions
    - Short and long breaks
- Track study duration
- Save sessions automatically

- Provide real-time countdown and animations

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f. Notifications System

- Request user permission for notifications
- Send reminders for tasks
- Notify users when a timer session ends
- Provide daily study reminders to maintain streaks

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g. Statistics and Analytics

- Calculate:
  - Daily and weekly study time
  - Study streak
  - Peak productivity hours
- Display data using charts and visual components
- Provide subject-wise study breakdown

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h. Testing and Debugging

- Test using iOS Simulator and real devices
- Debug UI and data issues
- Ensure smooth performance and reliability
- Validate data persistence and app behavior

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4. Project Qualifications

The project will be developed by a Computer Science student with experience in:

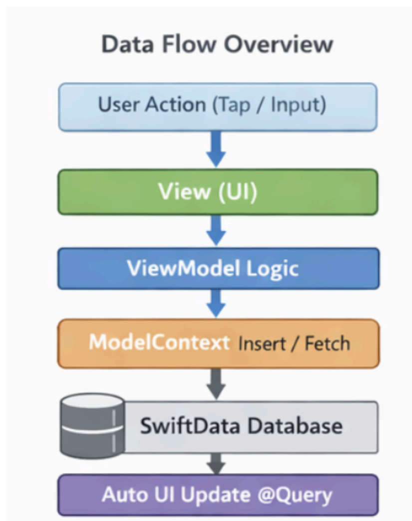
- Swift programming
- Object-oriented programming
- iOS app development using SwiftUI
- Database management (SwiftData/Core Data)
- Software development using Xcode

This project demonstrates the practical implementation of these skills in building a complete and functional mobile application.

5. Deliverables

By the end of the project, the following deliverables will be provided:

- A fully functional **StudyBuddy iOS application**
- Source code hosted on a **GitHub repository**
- A detailed **README file**
- Project documentation (design + architecture)
- Screenshots of application UI
- Final presentation and demonstration



6. Compliance and Data Usage

- The application will store data locally using SwiftData
- No personal or sensitive data will be collected or shared
- All project files and code will comply with course requirements
- The project will be submitted via GitHub as required

7. Appendix

Appendix A – GitHub Repository

Link to repository

Appendix B – Technologies Used

- Swift Programming Language
- SwiftUI Framework
- SwiftData (Local Database)
- Xcode IDE
- iOS SDK (iOS 17+)
- MVVM Architecture